The documentation and process conversion measures necessary to comply with this amendment shall be completed by 31 October 2000.

INCH-POUND

MIL-PRF-19500/548C
AMENDMENT 1
31 July 2000

PERFORMANCE SPECIFICATION SHEET

COUPLER, OPTO ELECTRONIC, SEMICONDUCTOR DEVICE, SOLID STATE
TYPES 4N47, 4N47A, 4N48, 4N48A, 4N49, 4N49A, 4N47U, 4N48U, AND 4N49U
JAN, JANTX, JANTXV, AND JANS

This amendment forms a part of MIL-PRF-19500/548C, dated 10 October 1997 and is approved for use by all Departments and Agencies of the Department of Defense.

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement pages	Page replaced
3	3
4	4

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA – NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 13, 19, 99

Preparing activity:
DLA - CC

(Project 5980-0022)

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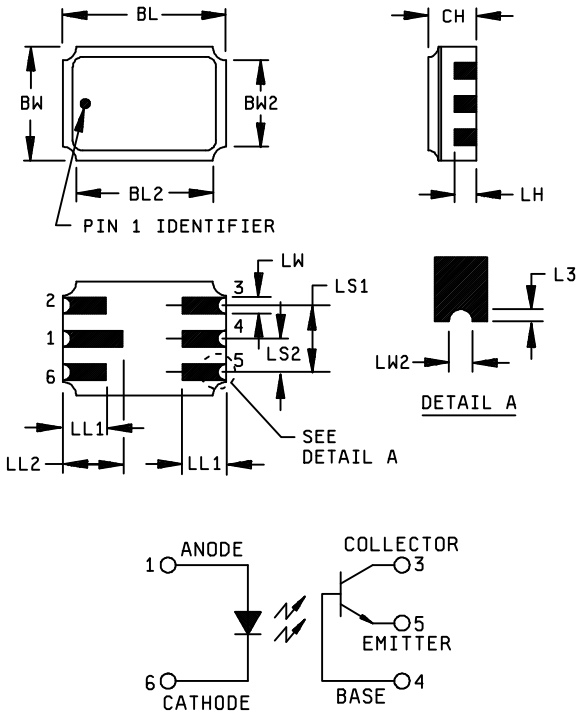
LTR	Dimensions			
	Inches		Millimeters	
	MIN	MAX	MIN	MAX
BL	0.240	0.250	6.10	6.35
BL2	----	0.250	----	6.35
BW	0.165	0.175	4.19	4.44
BW2	----	0.175	----	4.44
CH	0.066	0.080	1.68	2.03
LH	0.026	0.034	0.66	0.88
LL1	0.060	0.070	1.65	1.78
LL2	0.082	0.098	2.08	2.49
LL3	0.003	0.007	0.08	0.18
LS1	0.095	0.105	2.41	2.67
LS2	0.045	0.055	1.14	1.39
LW	0.022	0.028	0.56	0.71
LW2	0.006	0.022	0.15	0.56
N	6			
ND	3			
NE	0			



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension N: Number of terminals
4. Dimension ND/NE: Number of terminals per package side.

FIGURE 2. Dimensions and configuration (4N47U, 4N48U, and 4N49U).

Supersedes page 3 of MIL-PRF-19500/548C
of 10 October 1997.

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1.3.3 Total device ratings.

T_J	T_{STG}	V_{IO}
°C	°C	V dc
65 to +125	65 to +125	1,000 V (max)

1.4 Primary electrical characteristics at T_A = +25EC.

1.4.1 LED (input) characteristics at T_A = +25EC.

Limits	I_R V_R = 2 V dc	V_F I_F = 10 mA dc	I_F 1/ V_F = 1.5 V dc
	°A dc	V dc	mA dc
Minimum		0.8	10
Maximum	100	1.5	

1/ Under no circumstances, shall I_F exceed the maximum rating of the device during the test.

1.4.2 Phototransistor (output) characteristics.

Limits	I_C off V_{CB} = 20 V	I_{CB} off V_{CB} = 20 V	$V_{(BR)CEO}$ I_C = 1 mA	$V_{(BR)CBO}$ I_E = 100 µA	$V_{(BR)EBO}$ I_E = 100 µA	h_{FE} V_{CE} = 5 V dc, I_C = 10 mA dc, I_F = 0
	nA	nA	V dc	V dc	V dc	
Minimum			40	45	7	100
Maximum	100	10				

1.4.3 Coupled (transfer) characteristics.

Limits	R_{IO} at 1,000 V see 4.5.5	C_{IO} V = 0 f = 1 MHz see 4.5.5	Phototransistor mode switching (see figure 3)						Photodiode mode switching (see figure 3)	
			t_r			t_f			t_r	t_f
			4N47 4N47A 4N47U	4N48 4N48A4 N48U	4N49 4N49A 4N49U	4N47 4N47A 4N47U	4N48 4N48A 4N48U	4N49 4N49A 4N49U		
	ohms	pF	µs	µs	µs	µs	µs	µs	µs	µs
Minimum	10^{11}									
Maximum		5	20	20	25	20	20	25	3	3

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